3.3 – Measures of Spread – Worksheet
MDM4U
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1) Listed below are the life expectancies (in years) of men in 28 countries.

46.8	67.8	70.9	74.1	44.2	58.2	37.3	55.0	62.9	63.6
68.6	64.3	75.1	**75.9**	59.4	70.1	51.7	53.2	51.6	63.5
73.6	69.4	73.8	69.8	55.0	68.0	71.7	51.5		

a) What are the maximum and minimum values?

b) What is the range of life expectancies?

c) What is the median value?

d) What percent of the population is represented by the upper quartile?

e) What is the interquartile range?

2) Calculate the range, quartile values, and interquartile range of the data sets that follow.

a) $\{1, 3, 3, 5, 6, 6, 7, 7, 7, 8, 9\}$

b) $\{10, 14, 16, 20, 30, 50, 60, 68, 70, 90\}$

c) $\{1, 5, 5, 5, 7, 8, 9, 11, 12, 15, 15, 15, 16, 16, 18, 25, 29, 31\}$

d) $\{0, 2, 2, 3, 4, 5, 5, 5, 6, 6, 6, 6, 6, 6, 6, 6, 7, 8, 10, 12, 12, 18, 24, 28\}$

3) Calculate the standard deviation for each of the following sets of data. *(Use your calculator for a and d; b and c by hand)*

a) $\{0.2, 0.2, 0.3, 0.5, 0.5, 0.8, 0.8, 0.8, 0.9, 1.0, 1.0, 1.1, 1.2, 1.5\}$

b) $\{17, 18, 19, 20, 21, 22, 23, 24, 25, 26\}$

c) $\{5, 5, 5, 5, 5, 5, 6, 7\}$

d) $\{12\,943, 12\,942, 12\,947, 12\,941, 12\,939\}$

4) Use your calculator to determine the quartile values, interquartile range, and standard deviation of the salaries listed below.

Salary ($)	Frequency
28 000	4
30 000	6
32 000	7
34 000	4
36 000	2
38 000	1

5) Calculate the standard deviation for the data set in question 4 by hand.

6) Can the standard deviation of a set of data ever be equal to zero? Explain.

7) Four classes recorded their pulse rates.

Class A	63	78	79	75	73	72	62	75	63	77	77	65	70	69	80
Class B	72	66	73	80	74	75	64	68	67	70	70	69	69	74	74
Class C	68	75	78	73	75	68	71	78	65	67	63	69	59	68	79
Class D	78	75	76	76	79	78	78	76	74	81	78	76	79	74	76

Use the mean and standard deviation to determine which class has the lowest pulse rate and which class has the most consistent pulse rate. (Use your calculator)

8) Jaime has 20 min to get to her after school job. Despite her best efforts, she is frequently late and has recorded her travel time (in minutes) as {18, 20, 22, 27, 16, 23, 25, 26, 19, 28}. Her boss has told her that unless she shows more consistency, she will lose her job. Over the next two weeks, Jaime records her travel time: {22, 20, 22, 24, 24, 23, 25, 21, 19, 21}. Should Jaime lose her job? Use statistics to justify your answer.

Answers

1) a) min = 37.3; max = 75.9 b) 38.6 c) 63.95 d) 25% e) 17.7

2) a) range = 8, $Q_1 = 3, Q_2 = 6, Q_3 = 7, IQR = 4$

b) range = 80, $Q_1 = 16, Q_2 = 40, Q_3 = 68, IQR = 52$

c) range = 30, $Q_1 = 7, Q_2 = 13.5, Q_3 = 16, IQR = 9$

d) range – 28, $Q_1 = 5, Q_2 = 6, Q_3 = 10, IQR = 5$

3) a) 0.37 b) 2.87 c) 0.70 d) 2.65

4) range = 4000, $Q_1 = 30\ 000, Q_2 = 32\ 000, Q_3 = 34\ 000, IQR = 4\ 000$, mean = 31 750, $\sigma = 2665.36$

5) $\sigma = 2665.36$

6) Yes, if all numbers are the same

7) Class C has the lowest mean, therefore it has the lowest pulse rate. Class D has the lowest standard deviation, therefore, it has the most consistent pulse rate.

8) old mean = 22.4, old $\sigma = 3.9$, new mean = 22.1, new $\sigma = 1.8$; Jaime is not going to loser her job because she is more consistent now.